

Statistical methods –Data Science

Course Placement: It's a core course for students in Masters in Data Science Programme.

Course format: Three 50 minutes' lectures every week.

Lab: 2 hours every week

Course Content:

Data summarization: Data types, Data visualization and summarization with the help of tables and descriptive measures.

Probability: Statistical interpretation of probability, Bayes' theorem and its applications.

Standard probability models: Binomial, Poisson, Uniform, Exponential and Normal, Data generation from different models using Monte Carlo method.

Sampling: Concepts of population and sample, Sampling distributions of sample mean and sample proportion, Central limit theorems, Illustration of sampling distribution using Monte Carlo simulation.

statistical inference

Statistical estimation: Point and interval estimation, Methods of estimation (method of moments & maximum likelihood method)

Statistical testing of hypotheses: Null and alternative hypotheses, Significance testing using p-value, Standard parametric and non-parametric tests

Regression Analysis: Linear and Logistic regression

Time Series Analysis: Components of time series, Determination of Trend and Seasonal components

Special topics:

Resampling technique: Bootstrap

For all topics course notes will be provided by the instructor. In addition, the following books may be used as ready references.

References:

1. Statistics for Business & Economics – Anderson et al., Cengage Learning (2018)
2. Introduction to Data Science -- Rafael A. Irizarry (CRC Press but freely available in <https://leanpub.com/datasciencebook>)
3. An introduction to probability theory and its application (Volume -1) —William Feller (Wiley series in probability and statistics)
4. Statistical Inference – George Casella and Roger L. Berger (Cengage Learning India)
5. An introduction to the Bootstrap—Bradley Efron and Robert J. Tibshirani (Chapman Hall/CRC Press)

6. Introductory Statistics with R – Peter Daalgard, Springer (2008)

Assessment method: Quizzes, project presentation, and an end-term examination

Course Outcomes: After completion of this course, students will

- Learn the fundamental concepts and the basic tools of Data Analytics
- Learn the implementation of the analytics tools using R
- Acquire the skill to analyze data in a systematic and logical manner to answer important research questions arising in different disciplines, especially in the field of agriculture
- Learn how to attach a measure of uncertainty to the inference drawn from data
- Get a feel for data driven decision making

Lecture Schedule

Sl. No.	Description	No. of Lectures
1.	Data summarization	5
	Data types, Data visualization and summarization with the help of tables and the important descriptive measures.	
2.	Probability	5
	Concepts of probability (both objective and subjective). Statistical interpretation of probability, Bayes' theorem and its applications in solving real-life decision problems.	
3.	Standard probability distributions	10
	Standard probability models (Binomial, Poisson, Uniform, Exponential, Gaussian), Probability distributions of transforms of random variables, Data generation from probability distributions using Monte Carlo method.	
4.	Sampling	5
	Concepts of population and sample (both random and non-random), Parameter & Statistic, Probability distribution of a statistic (small sample & large sample), Central limit theorems, Illustration of sampling distribution using Monte Carlo simulation.	
5.	Statistical inference	5
	Statistical estimation: Point and interval estimation, Methods of estimation (method of moments & maximum likelihood method)	3

	Statistical testing of point null hypotheses: Null and alternative hypotheses, Significance testing using p-value, Some standard parametric and non-parametric tests	4
6.	Analysis of Variance	3
7.	Regression Analysis Linear Regression & Logistic Regression	10
7.	Time Series Analysis	5
9.	Bootstrap estimation of Bias and Standard Error	3